

Tresor MUGISHA

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PROFESSIONAL SUMMARY

I am a software Engineer specializing in backend development, with experience in building and deploying production-Grade SaaS products. I focus on creating reliable systems, writing maintainable code, and being a diligent team member.

EDUCATION

African Leadership University — Kigali, Rwanda

BSc Software Engineering | Expected Graduation: July 2028

- Software architecture, backend systems, and engineering leadership
- Database Systems and Data structures & Algorithms

PROFESSIONAL EXPERIENCE

Backend Developer (Remote) — Scrimba, Norway

2024 – 2025 (6 months)

Node.js, Express, TypeScript, MongoDB, REST APIs, Git

- Delivered production-grade REST APIs with JWT authentication, refresh token, and declarative route protection
- Engineered MongoDB schemas with indexing and constraints, eliminating orphaned records
- Built reusable middleware pipeline for logging, error normalization, and input sanitization across all endpoints
- Collaborated via GitHub PR workflows, delivering features on schedule with zero support tickets from frontend

Full Stack Developer & Lead — KLab Organization, Rwanda | Oct 2025 – Feb 2026

Kotlin, WebSockets, REST APIs, Git, GitHub, Trello

- Led backend development of *ShieldKids* (child safety app), owning architecture, sprint execution, and stakeholder demos
- Implemented persistent WebSockets for real-time location tracking and geofence breach alerts
- Architected RESTful APIs covering authentication, device registration, app controls, and screen time enforcement
- Managed all PR reviews, resolved integration conflicts, and delivered sprints from planning to demo

Backend Engineer (Remote) — MP Circle Ltd , South Africa | Mar 2026 – Apr 2026

NestJS, TypeScript, Docker, Swagger, GitHub

- Architected NestJS backend services with REST contracts, typed responses, and unified error handling
- Produced enterprise grade Swagger/OpenAPI docs enabling self-service client integration
- Conducted end-to-end API testing using Postman tool
- Troubleshoot Docker containerized environments to ensure builds in development stage

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+)

Backend: Node.js, NestJS, Express.js, REST API Design, WebSockets, JWT Authentication

Databases: PostgreSQL, MongoDB, MySQL, TypeORM, Schema Design, Query Optimisation

DevOps & Tools: Docker, Git, GitHub, Postman, Swagger, FCM

Frontend: React, Next.js, Tailwind CSS

KEY PROJECTS

IkiminaPass — ROSCA Platform for Rwanda – Backend & Project Lead

- Architected a full backend system in NestJS and PostgreSQL to digitise Rwanda's community savings groups (Ibimina)
- Integrating with payment gateway via IntouchPay for automated payment processing and contribution tracking
- Built USSD interface for feature phone access, enabling group management without access to our app
- Integrated Africa's Talking SMS and FCM for dual-channel notifications and real-time alerts

TheYouthContest — platform for debate organization | Fullstack

- Designed and built the full-stack infrastructure digitalizing debate championship processes
- Implemented robust Role-Based Access Control (RBAC) to ensure secure, role-specific data access.
- Built and deployed a clean frontend, optimizing React hooks for efficient data fetching.

Job Portal Full Stack Recruitment Platform | Backend & Project Lead

- Led full development lifecycle architecture, task distribution, version control, testing, and deployment
- Designed MongoDB schemas and implemented backend APIs for job listings, applications, and user role management
- Built authentication and protected route systems with role-based access for employers and applicants

BDifferent — E-commerce platform | Full-Stack | Personal

- Built backend CRUD APIs in Node.js and Express for product, user and category management with image upload support
- Architected optimized MongoDB schemas, establishing efficient data relationships to optimize product queries
- Integrated frontend and backend services using Axios, ensuring a scalable and decoupled architecture.
- Implemented schema validation middleware using Zod to ensure type-safe and secure API requests

INTERESTS

- Building real-time systems
- Contributing to open source
- Backend tooling automation
- Project Leading
- Devops

ACHIEVEMENTS

- KLab Training Compilation Certificate : [certificate](#)